Worked for 1m 35s

Yes — these results are genuinely informative. **v2.2 found a much better direction than the earlier “keep fine-tuning one adapter harder” approach.** But it also exposed two remaining limitations.

The central result is that `greedy_soup` **is the first candidate that adds a meaningful new capability while almost completely preserving the original student's strong specialized behavior.** Both selection policies independently choose it, and there is no policy disagreement.

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1. The main result: composition works

Here is the comparison I care about most:

Metric	Frozen original	Greedy soup	Change
v2.2 multidomain reasoning	0.000	**0.124**	**+12.4 pts**
v2.2 hard reasoning	0.000	**0.125**	**+12.5 pts**
Original Real-RAG procedure	**0.953**	**0.953**	**no loss**
Supported answers	0.626	**0.672**	**+4.5 pts**
Faithfulness	**0.973**	0.965	−0.8 pts
Original abstention	0.863	**1.000**	**+13.7 pts**
Social usability	1.000	**1.000**	no loss
Conflict handling	1.000	**1.000**	no loss

That is a very different pattern from v2.1/v2.1.2.

The preservation-guarded calculation gives:

benefit = 0.1325
preservation cost = 0.0128
net opportunity margin = +0.1197

So the improvement is not coming from simply exchanging one capability for another. capability_composition_v22_sele...

This is evidence for the hypothesis behind v2.2:

> The student has room for more capability, but **how the new capability is inserted and combined matters enormously**.

I would not interpret the earlier tradeoffs as proving that the model had reached a hard capacity ceiling anymore.

---

2. Track B may be the most important *training* result

Greedy soup is the best final candidate, but **Track B — anchored reasoning — may be the most scientifically important individual experiment**.

Compare unanchored versus anchored broad training:

	Track A broad	Track B anchored
Multidomain reasoning	**0.259**	**0.237**
Hard reasoning	**0.228**	**0.221**
Original procedure	**0.093**	**0.947**
Supported	0.077	**0.609**
Faithfulness	0.522	**0.874**
Abstention	0.727	**0.957**

Track B retained roughly **91% of Track A's multidomain score and ~97% of its hard-reasoning score**, yet almost completely prevented the catastrophic destruction of the original procedural capability.

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That is a huge result.

It strongly suggests that:

```
ordinary SFT
    ↓
learn new distribution
    ↓
overwrite old behavior
```

```
versus

new-task SFT
+ preservation replay
+ original-model KL anchor
    ↓
learn new distribution
while constraining destructive drift
```

works substantially better for this tiny student.

So I would make **behaviorally anchored training a default experimental primitive from now on**, not a special Track B experiment.

---

3. Plain broad training is definitely too destructive

Track A reached the highest raw multidomain score of **25.9%**, but look at what happened:

```
Original procedure:
95.3% → 9.3%

Supported RAG:
62.6% → 7.7%

Faithfulness:
97.3% → 52.2%
```

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That is catastrophic forgetting.

So the answer to:

> “Should we just make the general dataset even bigger and keep training?”

is now fairly clearly **no, not with ordinary SFT**.

More broad unanchored training would very plausibly continue increasing general reasoning while turning the student into a different model.

The problem is not merely the amount of data. It is **optimization interference**.

4. The specialist experiment worked — but in a revealing way

The five specialist adapters really did specialize.

The clearest example is abstention:

```
specialist_abstention

v2.2 abstention:       1.000
original abstention:   1.000
```

But it paid for that with:

```
procedure:     0.953 → 0.687
supported:     0.626 → 0.509
faithfulness:  0.973 → 0.802
```

capability_composition_v22_sele...

Likewise, the procedure specialist is the **only candidate that actually moved the hard v2.1.1 procedural challenge**:

```
0.000 → 0.060
```

But simultaneously:

```
original procedure:
0.953 → 0.213
```

That is fascinating.

It means we're not merely observing:

> “procedure training doesn't work.”

We are observing something closer to:

> **The procedure-challenge distribution teaches a behavior that conflicts with the original query-aware procedural behavior.**

That needs its own investigation.

I would not add another 10,000 procedural examples yet.

5. Naive weighted LoRA merging mostly failed

This experiment gave us a very clear answer.

The preregistered weighted merges produced:

```
balanced:            0.009 multidomain
grounding-guarded:   0.0005
reasoning-forward:   0.026
```

despite their constituent adapters individually having much stronger capabilities. CAPABILITY_COMPOSITION_V22_SELE...

So something like:

$$0.45\Delta_{\text{general}} + 0.20\Delta_{\text{grounding}} + \ldots$$

does **not** preserve the learned functions in a simple linear way.

This strongly suggests adapter deltas are interfering/canceling in parameter space.

That is a valuable negative result.

I would stop manually guessing static weights like:

```
0.5 / 0.2 / 0.1 / ...
```

for now.

---

6. But greedy composition succeeded

And this is why the previous result is particularly interesting.

Naive predefined merging failed.

Greedy selection succeeded.

The final greedy candidate has:

```
multidomain          12.42%
hard                 12.47%

procedure            95.33%
```

```
supported           67.17%
faithfulness        96.49%
abstention         100.00%
social             100.00%
conflict           100.00%
```

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So adapter composition itself is **not** fundamentally broken.

Instead:

> **Which deltas are combined, and at what strengths, matters much more than a hand-designed global recipe.**

The attached selection report doesn't include the exact accepted greedy ingredients/coefficients, so I would **not guess which adapters produced this result**. The greedy-search trace inside the v2.2 artifact root is now one of the most important things to inspect.

We should ask:

```
Which adapter entered first?
Which ones were rejected?
Which coefficients survived?
What candidate caused the largest jump?
Did adding another adapter undo it?
```

That can tell us substantially more than another training run.

---

7. The layer-separated experiment failed

The layer-separated merged model got only:

```
multidomain:  2.96%
```

while damaging:

```
procedure: 95.3% → 37.3%
supported: 62.6% → 27.0%
```

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So the hypothesis:

> “Just give general, RAG, procedure, abstention and social different non-overlapping layer bands”

is **not supported** by this experiment.

But I would not conclude that layer specialization is false.

The experiment assigned layer regions structurally. We did **not first prove which layers were actually best for each capability**.

The next layer experiment should therefore be:

```
CAPABILITY
   ↓
layer contribution / adapter importance scan
   ↓
actual high-contribution layers
   ↓
compare overlaps between capabilities
   ↓
only THEN design layer-aware adapters
```

In other words, **measure first; partition second**.

---

8. There is another result I don't want us to overlook

v2.2's broad reasoning training does **not transfer strongly to the old v2.1.2 reasoning benchmark**.

For example:

```
Track A
v2.2 multidomain: 25.9%
v2.1.2 overall:    0.38%

Track B
v2.2 multidomain: 23.7%
v2.1.2 overall:    0.30%
```

The frozen student's v2.1.2 score was already ~0.30%.

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Even the greedy soup reaches only:

```
v2.1.2 overall: 2.73%
hard:           12.86%
```

So I would be careful about saying:

> “The model has learned general reasoning.”

A more defensible statement is:

> **It has learned substantial capability on the v2.2 multidomain reasoning distribution, with some cross-benchmark transfer, but generalization across reasoning distributions is still weak.**

That matters a lot.

The next general dataset should therefore prioritize **structural diversity and OOD transfer**, not merely more domains.

---

9. And the original hard procedural challenge is still unsolved

Greedy soup:

```
v2.1.1 procedural challenge = 0.0
```

Track B:

```
0.0
```

All weighted merges:

```
0.0
```

Only the destructive procedure specialist moved it:

```
0.0 → 0.0603
```

capability_composition_v22_sele...

So the old hard collision problem remains open.

But this time I would **not make it the center of the entire project again**.

Instead, treat it as one specialized diagnostic axis.

What I would do next

There are two stages now.

Immediate step: run the sealed TEST

v2.2 has finally produced a candidate worth sealing.

Greedy soup was selected by both:

```
balanced
preservation_guarded
```

and policy sensitivity disagreement is false. capability_composition_v22_sele...

That is exactly when the sealed evaluation should be used.

Run the same workflow with the same original student arguments and existing output root:

PowerShell

```
.\scripts\phase_9_student_distillation\run_phase_9_capability_composition_v22.ps1 `
  -StudentModel "<SAME MODEL USED FOR THE DEV RUN>" `
  -StudentAdapter "<SAME ORIGINAL ADAPTER, IF USED>" `
  -OutputRoot ".\artifacts\phase_9_capability_composition_v22_lr1e4_s1200" `
  -ReuseExisting `
  -RunSealedTest `
  -RequireCuda
```

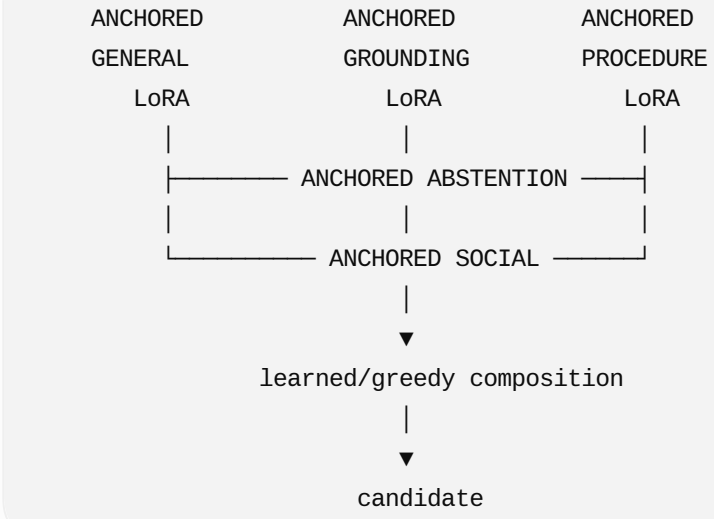
That should **not retrain** the candidates. It should freeze the DEV-selected `greedy_soup` and evaluate it on the sealed sets.

This is the right time to spend one sealed evaluation.

After sealed TEST: I would make v2.3 about Anchored Capability Composition

The next architecture almost writes itself from these results:





The crucial change is:

> **Every specialist gets the Track-B treatment.**

Not just general reasoning.

Because v2.2 showed that independent specialists learn their skill but move too far away from the original model.

So for each specialist:

$$L = L_{\text{specialist}} + \lambda_{\text{replay}} L_{\text{preservation}} + \lambda_{\text{KL}} D_{\text{KL}}(p_{\text{anchor}} \| p_{\text{student}})$$

Then run greedy composition over **anchored specialists**, not destructive specialists.

That is the most natural next hypothesis.

I would add two more experiments to that version

First, **adapter interference mapping**.

Before merging anything, calculate per-layer/per-module:

```
LoRA delta norm
cosine similarity between adapters
sign disagreement
overlapping high-magnitude directions
```

For example:

```
general Δ ↔ procedure Δ
general Δ ↔ grounding Δ
```

```
abstention Δ ↔ grounding Δ
```

If two adapters have strongly opposing deltas in the same projection matrices, we finally have a mechanistic explanation for why predetermined weighted merging failed.

Then composition can become **conflict-aware**, rather than blindly averaging.

Second, add an **oracle adapter-routing experiment**.

This isn't necessarily the production architecture. It's an upper-bound experiment:

```
question type known
       ↓
choose best specialist adapter
       ↓
answer
```

If oracle routing produces something like:

```
strong reasoning
+ strong original procedure
+ strong grounding
+ strong abstention
```

then we know the abilities **can coexist at the system level**, and merging is the bottleneck.

If oracle routing is still poor, then the specialist adapters themselves need improvement.

That would answer an extremely important question cheaply.

Dataset changes should now focus on generalization, not raw size

I would still enlarge the multidomain corpus, but v2.2 tells us **40,500 examples covering 27 domains is not sufficient evidence of truly general reasoning**.

The next dataset should vary the **representation of the reasoning problem** much more aggressively:

```
plain prose
tables
logs
code
configuration files
emails
chat
tickets
policy snippets
research abstracts
tool outputs
JSON
CSV-like rows
multi-document evidence
irrelevant distractors
longer state histories
cross-domain combinations
novel terminology
paraphrased instructions
unseen templates
```

And the most important addition should be a **structurally held-out generalization set**.

Not merely:

```
same generator
different numbers
```

but:

```
TRAIN:
some reasoning structures + surface forms

DEV:
different paraphrases

OOD DEV:
entirely held-out templates / compositions

SEALED TEST:
held-out templates + held-out domains/combinations
```

That would tell us whether the student learned an algorithm or memorized a family of prompt structures.

So my interpretation of v2.2 is actually quite positive

Earlier, the emerging story looked like:

> Tiny model learns one thing only by forgetting another.

v2.2 changes that.

We now have one candidate that simultaneously:

- preserves **95.3% original procedural accuracy**;
- raises supported RAG correctness;
- reaches **100% abstention** on the original evaluation;
- keeps social and conflict behavior perfect;
- loses less than one point of faithfulness;
- and acquires measurable new multidomain/hard reasoning capability.

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That is **not yet a broadly capable student**, and the challenge/generalization benchmarks tell us clearly where the gaps remain.

But it is the first strong evidence that the problem is not simply:

> “This student is too small.”

The much more interesting hypothesis now is:

> **The student has useful latent capacity, but ordinary fine-tuning causes destructive interference. Behavioral anchoring plus selective composition can recover some of that capacity without erasing the specialist.**

That is the direction I would now make the center of the research program: **preserve first, specialize independently, measure parameter conflict, and compose only compatible learned changes.**

Sources